

Table 1: NanoAOD 2023 datasets used in the analysis

Sample	NanoAOD dataset
Data (8 samples)	
Muon0_Run2023C_v1	/Muon0/Run2023C-22Sep2023_v1-v1/NANOAOD
Muon0_Run2023C_v2	/Muon0/Run2023C-22Sep2023_v2-v1/NANOAOD
Muon0_Run2023C_v3	/Muon0/Run2023C-22Sep2023_v3-v1/NANOAOD
Muon0_Run2023C_v4	/Muon0/Run2023C-22Sep2023_v4-v1/NANOAOD
Muon1_Run2023C_v1	/Muon1/Run2023C-22Sep2023_v1-v1/NANOAOD
Muon1_Run2023C_v2	/Muon1/Run2023C-22Sep2023_v2-v1/NANOAOD
Muon1_Run2023C_v3	/Muon1/Run2023C-22Sep2023_v3-v1/NANOAOD
Muon1_Run2023C_v4	/Muon1/Run2023C-22Sep2023_v4-v2/NANOAOD
Simulation (104 samples)	
GluGluHto2Mu_M120	/GluGluHto2Mu_M-120_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
GluGluHto2Mu	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
GluGluHto2Mu_amcatnlo	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
GluGluHto2Mu_MiNNLO	/GluGluHto2Mu_M-125_TuneCP5_13p6TeV_powhegMiNNLO-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
GluGluHto2Mu_tuneDown	/GluGluHto2Mu_M-125_TuneCP5Down_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
GluGluHto2Mu_tuneUp	/GluGluHto2Mu_M-125_TuneCP5Up_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
GluGluHto2Mu_M130	/GluGluHto2Mu_M-130_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
VBFHto2Mu_M125_powheg	/VBFHto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
VBFHto2Mu_m125_tuneCP5Up_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5Up_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
VBFHto2Mu_m125_tuneCP5Down_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5Down_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
ZH_Hto2Mu	/ZH_Hto2Mu_ZtoAll_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
WplusH_Hto2Mu	/WplusH_Hto2Mu_WtoAll_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
WminusH_Hto2Mu	/WminusH_Hto2Mu_WtoAll_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
TTH_Hto2Mu	/TTH_Hto2Mu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
VBFHto2Mu_M120_amcatnlo	/VBFH_Hto2Mu_M-120_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
VBFHto2Mu_m125_amcatnlo	/VBFH_Hto2Mu_M-125_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
VBFHto2Mu_M130_amcatnlo	/VBFH_Hto2Mu_M-130_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
DYto2Mu_MLL_105to160_amcatnloFXFX	/DYto2Mu-2Jets_MLL-105to160_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
DYto2L_M_10to50_amcatnloFXFX	/DYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v4/NANOADSIM
	/DYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14_ext1-v4/NANOADSIM
DYto2L_M_50_0J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_0J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v4/NANOADSIM
DYto2L_M_50_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v4/NANOADSIM
DYto2L_M_50_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v4/NANOADSIM
DYto2L_M_50_amcatnloFXFX	/DYto2L-2Jets_MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
DYto2L_M_50_PTL_40to100_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTL-40to100_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v1/NANOADSIM
DYto2L_M_50_PTL_40to100_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTL-40to100_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v1/NANOADSIM
DYto2L_M_50_PTL_100to200_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTL-100to200_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v1/NANOADSIM
DYto2L_M_50_PTL_100to200_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTL-100to200_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v1/NANOADSIM
DYto2L_M_50_PTL_200to400_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTL-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15_ext1-v2/NANOADSIM
	/DYto2L-2Jets_MLL-50_PTL-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
DYto2L_M_50_PTL_200to400_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTL-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15_ext1-v2/NANOADSIM
	/DYto2L-2Jets_MLL-50_PTL-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
DYto2L_M_50_PTL_400to600_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTL-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15_ext1-v2/NANOADSIM
	/DYto2L-2Jets_MLL-50_PTL-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
DYto2L_M_50_PTL_400to600_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTL-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15_ext1-v2/NANOADSIM
	/DYto2L-2Jets_MLL-50_PTL-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15_ext1-v2/NANOADSIM
DYto2L_M_50_PTL_600_1J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTL-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
	/DYto2L-2Jets_MLL-50_PTL-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15_ext1-v2/NANOADSIM
DYto2L_M_50_PTL_600_2J_amcatnloFXFX	/DYto2L-2Jets_MLL-50_PTL-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
	/DYto2L-2Jets_MLL-50_PTL-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15_ext1-v2/NANOADSIM
EWK_2L2J_madgraph_herwig	/EWK_2L2J_TuneCH3_13p6TeV_madgraph-herwig7/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
EWK_2Mu2J_MLL_105to160_herwig	/EWK_2Mu2J_Bin-MLL-105to160_TuneCH3_13p6TeV_madgraph-herwig7/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
EWK_2Mu2J_MLL_105to160_pythia	/EWK-2Mu2J_M2Mu-105to160_M2J-120_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
ggZH_Hto2B_Zto2Q	/ggZH_Hto2B_Zto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM
ggZH_Hto2B_Zto2L	/ggZH_Hto2B_Zto2L_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM

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Table 1: NanoAOD 2023 datasets used in the analysis (continued)

Sample	NanoAOD dataset
GluGluHto2B_M125	/GluGluHto2B_M-125_TuneCP5_13p6TeV_powheg-minlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
GluGluHto2Tau_Un-correlatedDecay_Filtered	/GluGluHto2TauUn-correlatedDecay_Filtered_M-125_CP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
GluGluHto2Tau_Un-correlatedDecay_UnFiltered	/GluGluHto2TauUn-correlatedDecay_M-125_CP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
GluGluHto2Wto2L2Nu_M125	/GluGluHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
TTHto2B_M125	/TTHto2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM
TTHtoNon2B_M125	/TTHtoNon2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
VBFHto2B_M125	/VBFHto2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM
VBFHto2Tau_M125	/VBFHtoTauTau_M125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
VBFHto2Tau_Un-correlatedDecay_UnFiltered	/VBFHto2TauUn-correlatedDecay_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
VBFHto2Tau_Un-correlatedDecay_Filtered	/VBFHto2TauUn-correlatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
VBFHto2Wto2L2Nu_M125	/VBFHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
WminusH_Hto2B_WtoLnu	/WminusH_Hto2B_WtoLnu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
WplusH_Hto2B_WtoLnu	/WplusH_Hto2B_WtoLnu_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
WminusHto2Tau_Un-correlatedDecay_Filtered	/WminusHto2TauUn-correlatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
WminusHto2Tau_Un-correlatedDecay_UnFiltered	/WminusHto2TauUn-correlatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
WplusHto2Tau_Un-correlatedDecay_Filtered	/WplusHto2TauUn-correlatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
WplusHto2Tau_Un-correlatedDecay_UnFiltered	/WplusHto2TauUn-correlatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
ZH_Hto2B_Zto2L	/ZH_Hto2B_Zto2L_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
ZH_Hto2B_Zto2Q	/ZH_Hto2B_Zto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
WminusH_Hto2B_Wto2Q	/WminusH_Hto2B_Wto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM
WplusH_Hto2B_Wto2Q	/WplusH_Hto2B_Wto2Q_M-125_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
TbarBtoLminusNuB_s_channel_4FS	/TbarBtoLminusNuB-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
TBbartoLplusNuBbar_s_channel_4FS	/TBbartoLplusNuBbar-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
TbarWplusto2L2Nu	/TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v4/NANOADSIM
TbarWplusto4Q	/TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v4/NANOADSIM
TbarWplustoLnu2Q	/TbarWplustoLnu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v6/NANOADSIM
Twminusto2L2Nu	/Twminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
Twminusto4Q	/Twminusto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
TwminustoLnu2Q	/TwminustoLnu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
TTto2L2Nu	/TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
TTto4Q	/TTto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
TTtoLnu2Q	/TTtoLnu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
TTWH	/TTWH_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v3/NANOADSIM
TTWW	/TTWW_TuneCP5_13p6TeV_madgraph-madspin-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
TbarBQ_t_channel_4FS	/TbarBQ_t-channel_4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
TBbarQ_t_channel_4FS	/TBbarQ_t-channel_4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
TTZH	/TTZH_TuneCP5_13p6TeV_madgraph-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v3/NANOADSIM
TTZZ	/TTZZ_TuneCP5_13p6TeV_madgraph-madspin-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v4/NANOADSIM
WtoLnu_1J_madgraphMLM	/WtoLnu-4Jets_1J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM
WtoLnu_2J_amcatnloFXFX	/WtoLnu-2Jets_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
WtoLnu_2J_madgraphMLM	/WtoLnu-4Jets_2J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM
WtoLnu_3J_madgraphMLM	/WtoLnu-4Jets_3J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM
WtoLnu_4J_madgraphMLM	/WtoLnu-4Jets_4J_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM
WtoLnu_0J_amcatnloFXFX	/WtoLnu-2Jets_0J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM
WtoLnu_1J_amcatnloFXFX	/WtoLnu-2Jets_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
WtoLnu_amcatnloFXFX	/WtoLnu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
WtoLnu_madgraphMLM	/WtoLnu-4Jets_TuneCP5_13p6TeV_madgraphMLM-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM
WW	/WW_TuneCP5_13p6TeV_pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
WZ	/WZ_TuneCP5_13p6TeV_pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
ZZ	/ZZ_TuneCP5_13p6TeV_pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
WWto2L2Nu_powheg	/WWto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v4/NANOADSIM
WWto4Q_powheg	/WWto4Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v4/NANOADSIM
WWtoLnu2Q_powheg	/WWtoLnu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM
WZto2L2Q_powheg	/WZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM

Continued on next page

Table 1: NanoAOD 2023 datasets used in the analysis (continued)

Sample	NanoAOD dataset
WZto3LNU_powheg	/WZto3LNU_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v2/NANOADSIM
	/WZto3LNU_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15_ext1-v2/NANOADSIM
WZtoLNU2Q_powheg	/WZtoLNU2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
ZZto2L2Nu_powheg	/ZZto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM
ZZto2L2Q_powheg	/ZZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM
ZZto2Nu2Q_powheg	/ZZto2Nu2Q_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM
ZZto4L_powheg	/ZZto4L_TuneCP5_13p6TeV_powheg-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v3/NANOADSIM
WWW_4F	/WWW_4F_TuneCP5_13p6TeV_amcatnlo-madspin-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
WWZ_4F	/WWZ_4F_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
WZZ	/WZZ_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
ZZZ	/ZZZ_TuneCP5_13p6TeV_amcatnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v14-v2/NANOADSIM
ZHto2TauUncorrelatedDecay_Filtered	/ZHto2TauUncorrelatedDecay_Filtered_M-125_CP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v5/NANOADSIM
ZHto2TauUncorrelatedDecay_UnFiltered	/ZHto2TauUncorrelatedDecay_M-125_CP5_13p6TeV_powheg-minnlo-pythia8/Run3Summer23NanoAODv12-130X_mcRun3_2023_realistic_v15-v5/NANOADSIM